

Zepto E-Commerce Inventory SQL Analysis — Business Problem Statement

1. Understanding the Business Problem

The main purpose of this project is to analyse Zepto's e-commerce inventory data and understand product performance, pricing behaviour, stock availability, and inventory-related business challenges.

The company manages a large product catalogue containing multiple categories, SKUs, pricing variations, discounts, and inventory levels. The business requires data-driven insights to improve inventory management, pricing decisions, and product availability.

2. Identifying Inventory and Product Performance Patterns

The company wants to understand how different products and categories perform within its inventory system. This analysis focuses on identifying patterns related to:

- Product category distribution
- Stock availability and out-of-stock products
- Discount percentages and pricing strategies
- Product demand opportunities
- High-value products and revenue potential
- Product weight and pricing efficiency

These insights will help the business identify improvement areas and optimize inventory operations.

3. Purpose of Data Analysis

By analysing Zepto's inventory dataset, this project aims to generate meaningful insights that can support business decisions related to:

- Inventory optimization
- Product pricing strategies
- Discount effectiveness
- Category performance evaluation
- Revenue opportunity identification
- Improving product availability

The analysis will help transform raw inventory data into useful business information using SQL.

4. Database Creation and Data Preparation Using SQL

The first stage of the project involves preparing the raw inventory dataset for analysis. This includes:

- Creating a structured SQL database and inventory table
- Defining appropriate data types for product attributes
- Importing raw CSV inventory data into the database
- Exploring dataset structure and records
- Identifying missing values and inconsistent data
- Removing invalid records such as products with zero or incorrect prices
- Converting pricing values into a consistent format for analysis

The cleaned dataset will be used for accurate SQL-based business analysis.

5. Data Analysis Using SQL

After data cleaning, SQL queries will be developed to answer important business questions and generate insights, including:

- Identifying the top products with the highest discount percentages
- Finding expensive products that are currently out of stock
- Estimating potential revenue across product categories
- Comparing average discounts between different categories
- Identifying products with better price-to-weight value
- Grouping products based on weight categories
- Analysing total inventory quantity and distribution by category

The SQL analysis will help stakeholders understand inventory performance and identify business opportunities.

6. Business Insights and Recommendations

The final analysis will summarise key findings from the SQL queries and provide practical recommendations.

The recommendations will focus on:

- Improving stock management for high-value products
- Optimizing discount strategies
- Monitoring category-level inventory performance
- Identifying products with higher revenue potential
- Improving pricing decisions based on customer value

These insights will support better inventory planning and operational decision-making.

7. Project Documentation and SQL Repository

All project materials, including SQL scripts, database creation queries, data cleaning queries, and business analysis queries, will be organised in a structured repository.

Proper documentation will ensure the project is easy to understand, review, and maintain as a complete SQL data analytics project.